

Homework 2

Solve the following problems justifying all your answers. You should not use a graph calculator to justify any answer.

1. Prove that the following functions are one-to-one and give a formula for their inverse.

(a) $f(x) = \frac{x^2 + x + 1}{x}, x \geq 1.$

(b) $g(x) = \begin{cases} -\frac{x-1}{x-3} + \frac{7}{3}, & x \leq 0, \\ e^{x+\sqrt{x}} + 1, & x > 0. \end{cases}$

2. Graph the following functions, not by plotting points, but by starting with the graph of one of the standard functions (like the square root function or the cubic function) and applying an appropriate transformation (for example shift, contraction/stretch, reflection):

(a) $f(x) = -\sqrt{1 + \frac{x}{3}},$

(b) $g(x) = (1 - x)^3 + 2$

3. Sketch the graph of a function f that satisfies all of the following conditions:

- $\lim_{x \rightarrow -4^+} f(x) = -\infty.$
- $\lim_{x \rightarrow -2^-} f(x) = \infty.$
- $\lim_{x \rightarrow -2^+} f(x) = -\infty.$
- $f(-2) = 1.$
- f is increasing in $(-4, -2) \cup (-2, -1).$
- $-1 \notin \text{Dom}(f).$
- $\lim_{x \rightarrow -1} f(x) = 2.$
- f is decreasing in $(-1, 0).$
- $f(0) = -2.$
- $\lim_{x \rightarrow 0^-} f(x) = -\infty.$
- $\lim_{x \rightarrow 0^+} f(x) = -2.$
- f is constant in $(0, 1).$
- $f(1) = 0.$
- f is constant in $(1, 4].$
- $\lim_{x \rightarrow 2^+} f(x) = 2.$

4. Find the values of a and b such that $\lim_{x \rightarrow 2} f(x)$ and $\lim_{x \rightarrow 3} f(x)$ exist:

$$f(x) = \begin{cases} \frac{x^2 - 4}{x - 2} & x < 2, \\ ax^2 - bx + 3 & 2 \leq x < 3, \\ 2x - a + b & x \geq 3. \end{cases}$$